

Homework 10: Theory-Based Inference

STAT 141, Week 12

Assigned: Thursday, November 19th

Due: Thursday, December 3rd at 9:00am on Gradescope as a PDF

In this homework assignment you will:

1. Make theory-based inferences for summary statistics in R
2. Assess data to determine whether assumptions for theory-based inference are met
3. Compare theory-based inference methods with simulation-based methods

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Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```
# Load the necessary packages
library(tidyverse)
library(infer)
```

Exercises

Exercise 1: How does the proportion affect the margin of error?

Imagine you've set out to survey 1000 people on two questions: Do you have brown eyes? Are you left-handed? In both cases, our parameter of interest is a single proportion. Since both sample proportions from our surveys were calculated from the same sample size, they should have the same margin of error, right? **Nope!** While the margin of error does change with sample size, it is also affected by the population proportion.

Think back to the formula for the standard error:

$$SE = \sqrt{\frac{p(1-p)}{n}}$$

This is then used in the formula for the margin of error for a 95% confidence interval:

$$ME = 1.96 \cdot SE = 1.96 \cdot \sqrt{\frac{p(1-p)}{n}}$$

Since the population proportion p is in this ME formula, it should make sense that the margin of error is in some way dependent on the population proportion. We can visualize this relationship by creating a plot of ME vs. p .

Since sample size is irrelevant to this discussion, let's just set it to some value ($n = 1000$) and use this value in the following calculations:

```
n <- 1000
```

The first step to visualize this is to make a variable p that is a sequence from 0 to 1 with each number incremented by 0.01. We can then create a vector of the margin of error (me) associated with each of these values of p using the familiar approximate formula ($ME = 2 \times SE$).

```
p <- seq(0, 1, 0.01)
me <- 2 * sqrt(p * (1 - p)/n)
```

Lastly, we combine these vectors into a dataframe so that we can visualize it either using `geom_point()` or `geom_line()` (or both!).

```
dd <- data.frame(p = p, me = me)
```

a. Use `ggplot` to plot the relationship between p and me . [Code only for this part]

b. Describe the relationship shown in the plot you created above. **Then, state for which value of p the margin of error maximized.** [Text only for this part]

c. Based on what we've learned here, which of the surveys would you expect to have a smaller standard error: the one estimating the proportion of brown-eyed adults, or the one estimating the proportion of left-handed adults? [Text only for this part]

(*Note:* An estimated 45% of US adults have brown eyes, and an estimated 10%-12% are left-handed.)

Exercise 2: Success-failure condition

We emphasized in lecture that you must always check conditions before making inferences. For inference on proportions, the sample proportion can be assumed to be nearly normal if it is based upon a random sample of independent observations and if both $np \geq 10$ and $n(1 - p) \geq 10$. This rule of thumb is easy enough to follow, but it makes one wonder: what's so special about the number 10?

The short answer is: nothing! You could argue that we would be fine with 9 or that we really should be using 11. What is the “best” value for such a rule of thumb is, at least to some degree, arbitrary (similar to how $\alpha = 0.05$ is relatively arbitrary). However, we often observe that when np and $n(1 - p)$ reaches 10, the sampling distribution is sufficiently normal to use confidence intervals and hypothesis tests that are based on that approximation.

We can investigate the interplay between n and p and the shape of the sampling distribution of $\hat{p}$ by using simulations. We've written a simple app to do this in R. To use it, open the `app.R` file in this homework folder, and without changing anything in the file, click “Run App” (towards the right corner on the bar where you normally see “Render”).

Play around with the app to investigate how the shape, center, and spread of the distribution of $\hat{p}$ changes as n and p changes.

a. Describe the sampling distribution of sample proportions at $n = 300$ and $p = 0.1$. Be sure to note the center, spread, and shape (eyeball the relevant numeric summary statistics for center and spread). You might want to adjust min and max for the x-axis for a better view of the distribution. [Text only for this question]

b. Keep n constant at $n = 100$, change p . How does the shape, center, and spread of the sampling distribution vary as p changes? Remember to update the min and max settings for the x-axis if you adjusted them. [Text only for this question]

c. Now fix p at 0.05 and change n . How does n appear to affect the distribution of $\hat{p}$? [Text only for this question]

Exercise 3: Theory-Based Inference for Difference in Proportions

This exercise continues our work with the atheism survey data from Lab 10, now focusing on inference for a **difference in proportions**, rather than inference for a single proportion. Load the data set again by running the chunk below. You can also revisit [the report from the poll here](#).

```
atheism <- read.csv("data/atheism.csv")
```

a. Find the sample proportion of atheists in Canada in 2012. Recalling that we found 50 out of 1002 respondents reported being atheists in the US in 2012, calculate the estimated *difference in proportions* of atheists in the US and Canada. [Code + text to distinguish your answers for this question]

b. Perform a hypothesis test (with $\alpha = 0.01$) to determine whether the true proportion of atheists in Canada and in the United States differed in 2012. Use *theory-based inference*, and be sure to write out:

- i. The hypotheses you're testing
- ii. Whether conditions for theory-based inference have been met
- iii. Your test statistic
- iv. Your p-value
- v. Your final conclusion

You should use a combination of code and text to show and explain your work.

c. If we wanted to do a simulation-based version of the hypothesis test from (b), what would we change? Be precise about describing the simulation aspect of this, but you do not need to write code to do it.

d. Using theory-based inference methods, construct a 90% confidence interval for this difference in proportions using *percentiles*. [Only code is needed for this question]

e. If we wanted to create a simulation-based version of the confidence interval from (d), what would we change? Be precise about describing the simulation aspect of this, but you do not need to write code to do it.

Exercise 4: Inference with a Single Mean

Let's now return to the combined [Opportunity Insights dataset](#) and college endowment data set from Homework 5. For the purpose of this lab, let's proceed as if this is a random sample of elite colleges in the U.S, and that we want to make inferences about the *average* endowment assets per student in 2016 across all elite colleges.

```
load("data/colleges_endow.Rdata")
```

a. Create a histogram of the `fy2016_endowment_pc` column, which contains data on the monetary value of endowment assets per student (in 2016), measured in USD. **Does it appear approximately Normal? Even if not, what allows us to still proceed with theory-based inference for the average endowment value?** [Text and code for this part]

b. Using theory-based inference and *the standard error method*, calculate a 90% confidence interval for the *average endowment assets per student*. Interpret the interval **in context**. [Text and code for this part]

c. Create a bootstrap distribution of the sample mean using our sample of `fy2016_endowment_pc` (isolated in the chunk below as `endow_sample`). **Does this distribution appear to be approximately Normal? Use the percentile method to generate a 90% confidence interval and compare to your previous result in part (b).** [Text and code for this part]

```
endow_sample <- data.frame(fy2016_endowment_pc = colleges$fy2016_endowment_pc)

### Your work below
```

Exercise 5: Anchoring Experiment

During the Lab 2, we collected data to see if our class exhibits an “anchoring effect”. The anchoring effect is a cognitive bias in which an individual’s decisions in an uncertain situation are reliant on the first piece of information received. Half of you received the number $x = 82$ and the other half the number $x = 531$, and then you took a survey that asked you to guess how many dog breeds there are in the world. The numbers were handed out at random. Recall that we’ve qualitatively investigated the following question: Does the number received serve as an anchor value for answering the second question? In other words, does the number seem to serve as a starting point when estimating the number of dog breeds in the world? Remember that last year, students in 141 participated in the same experiment. In this exercise we’ll work again with the pooled data across all semesters.

In previous exercises and activities, we’ve looked for evidence of an anchoring bias in an exploratory manner, without using formal statistical inference procedures. But now, we have the tools we need to conduct formal statistical inference!

```
# Load and clean the data
dogs <- read.csv("data/dogs_cleaned.csv")
dogs <- dogs %>%
  mutate(treatment = factor(treatment)) %>%
  select(treatment, n_breeds)
head(dogs)
```

	treatment	n_breeds
1	82	200
2	82	20
3	82	350
4	531	800
5	531	274
6	531	75

a. Let’s start by addressing the question: “Did this experiment demonstrate that an anchoring effect exists?” To answer this question, **(1)** are we looking to perform inference on a single proportion, a difference in proportions, a single mean, or a difference in means? And **(2)** why is a hypothesis test more relevant than a confidence interval for this specific question? [Text only for this question]

b. What are the two implied populations of this investigation? What are the two samples?
[Text only for this question]

c. State null and alternative hypotheses that could answer the research question based on the type of data collected in this investigation. Keep in mind that if an anchoring effect exists, we do expect higher priming numbers to result in higher responses. [Text only for this question]

d. Write down the conditions for using theory-based methods to carry out this hypothesis test. Then, explain why you think they have or have not been met in this situation. [Text only for this question]

e. Regardless of your answers to the previous question, we will use theory-based methods to assess the hypotheses in part (c). Compute the relevant *standardized test statistic*. What distribution does the test statistic follow, if the null hypothesis is true? [Text and code for this question]

f. Using theory-based methods, compute the p-value for the test statistic you found in the previous part, given your hypotheses from part (c). Then make a conclusion at the $\alpha = 0.05$ significance level, **in context**. [Text and code for this question]

g. Create a 95% confidence interval for our test statistic *using the SE method* for theory-based inference. [Code only for this question]
